TP09: Queue and Stack

- The objective of this Lab is
- To practice writing program in C++ using Queue
- Deadline: **1 week**
- Submit to Moodle
    - A pdf report
    - Source codes (No need to zip)
    - Files to submit: 1) A pdf report, 2) A zip source code file

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1. Create an implementation of Stack as Array. Create the variable and functions as below
    - const int SIZE = 3;
    - int stack[SIZE];
    - int top = -1;
    - bool isEmpty();
    - bool isFull();
    - void push(int data);
    - void pop();
    - void display();

2. Create an implementation of Stack as Linked list. Make them as header files. Hint: You can reuse the Single Linked List code and rename the operations and variable names to the terms used by Stack. (Note: The header files: MyStack.h)
    - Stack* createStack();
    - void push(Stack* s, int data);
    - void pop(Stack* s);
    - void display(Stack* s);
3. Create a function that removes all elements from the Stack in Exercise 2.
4. Create a function to delete the stack in Exercise 2 and its elements.
5. Create a function that removes all even elements from the Stack in exercise 2.
    - Original stack: 1 2 3 4 5
    - After removing all even numbers from the stack: 1 3 5
6. Create a function that finds the maximum element of a Stack. in exercise 2.
7. Create a function that finds the second highest element of a Stack in exercise 2.
8. Create a function that removes all duplicate elements from a Stack in exercise 2.
9. Create a function that display element data in the reverse order and remove from the Stack .
10. Create a function that display element data in the reverse order without remove from the Stack.